Growth and Stem-Cell Data: Evidence Before Claims

I can interpret growth and stem-cell data, calculate changes or proportions, and evaluate application claims using evidence and biological limitations.

DESCRIBE DATA
values + units

CALCULATE
show working

EVALUATE CLAIM
benefit + limit

Open the original interactive lesson

What success looks like

- Describe a growth pattern from axes, values and units without diagnosing from one data point.
- Calculate a change, percentage change or cell proportion with shown working.
- Apply stem-cell and differentiation knowledge to evaluate a claim, separating potential benefit from evidence limits and uncertainty.

Retrieve and reconnect

- State one limitation of interpreting a single growth measurement.
- Write the equation for percentage change.
- Compare embryonic and adult tissue stem-cell potential.
- Lead-in: distinguish evidence of differentiation from evidence of patient benefit.

A fictional culture study reports that 72 of 90 cells show the intended marker. Which response is scientifically strongest?

Think first. Point, say, sign or write your choice. Give one reason.

A. The treatment is guaranteed to cure 80% of patients.

B. 80% showed the marker in this culture; clinical safety and benefit still require separate evidence.

C. All 90 cells differentiated because 72 is close to 90.

Look closely · what is the evidence?

$72 \div 90 = 0.80$

$\times\ 100 = 80\%$

Proportion

$72 \div 90 = 0.80$

$\times\ 100 = 80\%$

Percentage

80% of cells showed the marker in a dish

≠ 80% of patients would be cured

bounded claim: the treatment increased marker-positive cells in this sample

Bounded claim

Words we will use

- percentile — position in a reference distribution
- efficacy — how well an intervention works
- inference — reasoned interpretation

Watch the model

- For a fictional growth chart, read age, measurement and unit, then describe direction and change using values rather than labels such as normal or abnormal.
- Calculate a change or percentage change and check the denominator. If a percentile is supplied, describe relative position and pattern over time without treating it as a diagnosis.
- Separate layers: the chart directly records organism measurements; mitosis, elongation and differentiation are biological explanations that require additional evidence.

A fictional pupil's height follows approximately the same percentile line at three time points. Which conclusion is justified by this chart alone?

A. The measurements show a broadly consistent relative growth pattern across these time points.

B. Every cell divided at the same rate and no health check could ever be needed.

C. The chart proves exactly which genes were active in each tissue.

Commit to an answer. Show the evidence you used. Listen, then revise if needed.

Build the explanation

- For a fictional stem-cell dataset, identify groups, sample sizes, measured outcome and time point before calculating a proportion or difference.
- Use the result to make a bounded claim: the intervention increased or decreased the measured outcome under the study conditions.
- Evaluate application by pairing a potential biological benefit with at least one limitation such as small sample, short follow-up, missing control, adverse effects or an indirect marker.

Find and repair the misconception

Application glitch: the group with the larger average must prove the treatment works for every person and has no risks.

A. Keep it: one larger mean establishes universal safety and efficacy.

B. Repair it: quantify the difference, check variation and controls, then state benefits and limitations before judging.

C. Repair it: ignore all quantitative evidence and decide from opinion alone.

Claim-strength evidence triage

Route each statement to directly supported, reasonable inference or unsupported overclaim before composing the evaluation.

- Read the dataset and one candidate statement.
- Commit it to the evidence strength it deserves.
- Repair it by adding a value, biological link or limitation.
- Freeze a conclusion that is useful without becoming diagnostic or promotional.

Use the numbered cards in your pupil pack. Take turns to choose, justify and check. You may direct a partner to move a card.

Choose a group · defend your placement

- DIRECTLY SUPPORTED
- REASONABLE INFERENCE
- UNSUPPORTED OVERCLAIM

State one direct result, one justified inference and one claim the dataset cannot support.

Your independent mission

Complete a GCSE application pack containing a growth-chart interpretation, two shown calculations and an evidence-led evaluation of a fictional stem-cell claim.

Record: A value-and-unit growth description, shown calculation, stem-cell data comparison, biological explanation, limitation and supported judgement ready to feed W11 ethics work.

Choose your support and challenge

- Supported: Read two graph values, calculate one change and one worked percentage, then select and justify a bounded conclusion.
- Standard: Interpret the growth pattern, calculate change and cell percentage, then evaluate the fictional application using one benefit and two evidence limitations.
- Stretch: Compare groups using quantitative evidence, examine variation or control quality, and reach a conditional judgement that separates efficacy, safety and ethics.

Show what you know

Use one value to support a stem-cell application claim and one limitation to bound it.

Point to, read or explain the evidence you made. Choose one next step after the adult has listened.